

Comparison of RNN, LSTM, and GRU for Sequence Classification

Problem Statement: Implement and compare RNN, LSTM, and GRU models for sequence classification, and analyze their performance using appropriate evaluation metrics.

Objective

- Implement a basic RNN
- Implement an LSTM
- Implement a GRU
- Train all models on the same dataset (IMDB movie reviews) under identical conditions
- Compare their performance using accuracy, precision, recall, F1-score, training time, and parameter counts
- Analyze the strengths and weaknesses of each architecture

Dataset: IMDB movie review dataset (binary sentiment classification: 0 = Negative, 1 = Positive), loaded via `tf.keras.datasets.imdb`.

Algorithm

1. Load the IMDB movie review dataset.
2. Explore the dataset and analyze sequence lengths.
3. Split the training data into training and validation sets.
4. Convert reviews into fixed-length padded sequences.
5. Create TensorFlow data pipelines.
6. Build an RNN model using an embedding layer and SimpleRNN.
7. Train and evaluate the RNN model.
8. Build and train an LSTM model using the same data and training settings.
9. Build and train a GRU model using the same data and training settings.
10. Calculate accuracy, precision, recall, F1-score, and test loss.
11. Generate confusion matrices and training curves.
12. Compare training time and model parameters.
13. Identify the best-performing model.
14. Analyze the results and conclude which recurrent architecture is most suitable for the task.

1. Import Libraries and Environment Setup

```
# Core deep learning framework
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers, Sequential
from tensorflow.keras.layers import Embedding, SimpleRNN, LSTM, GRU,
```

```

Dense
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping

# Data handling and numerical computation
import numpy as np
import pandas as pd

# Plotting
import matplotlib.pyplot as plt
import seaborn as sns

# Evaluation metrics
from sklearn.model_selection import train_test_split
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score,
    f1_score, confusion_matrix, classification_report
)

# Misc utilities
import time
import random
import os

# Display the TensorFlow version being used
print("TensorFlow version:", tf.__version__)

# Check whether a GPU is available and print the selected device
gpu_devices = tf.config.list_physical_devices("GPU")
if gpu_devices:
    print("GPU is available:", gpu_devices)
else:
    print("No GPU found. Running on CPU.")

TensorFlow version: 2.20.0
No GPU found. Running on CPU.

# -----
# Reproducibility: fix random seeds across all relevant libraries
# -----
SEED = 42

# Fix Python's built-in random module seed
random.seed(SEED)

# Fix NumPy's random seed
np.random.seed(SEED)

# Fix TensorFlow's random seed (covers Keras layer initialization too)
tf.random.set_seed(SEED)

```

```
# Fix the PYTHONHASHSEED environment variable for extra determinism
os.environ["PYTHONHASHSEED"] = str(SEED)

print(f"All random seeds fixed to {SEED} for reproducibility.")

All random seeds fixed to 42 for reproducibility.
```

2. Configuration

```
# -----
# Central configuration cell - change these values to experiment
# -----
VOCAB_SIZE = 10000      # Number of most frequent words to keep
MAX_LEN = 200           # Fixed sequence length after
padding/truncation
EMBEDDING_DIM = 64      # Dimensionality of the embedding vectors
RNN_UNITS = 64         # Number of units in the recurrent layer
DENSE_UNITS = 32       # Number of units in the intermediate dense
layer
BATCH_SIZE = 64        # Batch size used for training/evaluation
EPOCHS = 5             # Maximum number of training epochs
LEARNING_RATE = 0.001  # Learning rate for the Adam optimizer

print("Configuration loaded:")
print(f" VOCAB_SIZE      = {VOCAB_SIZE}")
print(f" MAX_LEN          = {MAX_LEN}")
print(f" EMBEDDING_DIM      = {EMBEDDING_DIM}")
print(f" RNN_UNITS          = {RNN_UNITS}")
print(f" DENSE_UNITS        = {DENSE_UNITS}")
print(f" BATCH_SIZE         = {BATCH_SIZE}")
print(f" EPOCHS             = {EPOCHS}")
print(f" LEARNING_RATE      = {LEARNING_RATE}")

Configuration loaded:
VOCAB_SIZE      = 10000
MAX_LEN         = 200
EMBEDDING_DIM   = 64
RNN_UNITS       = 64
DENSE_UNITS     = 32
BATCH_SIZE      = 64
EPOCHS          = 5
LEARNING_RATE   = 0.001
```

3. Load IMDB Dataset

```
# Load the IMDB movie review dataset
# Limit the vocabulary to the VOCAB_SIZE most frequently used words
(x_train_full, y_train_full), (x_test, y_test) =
keras.datasets.imdb.load_data(num_words=VOCAB_SIZE)
```

```

# Print basic dataset information
print("Training samples:", len(x_train_full))
print("Testing samples:", len(x_test))
print("Number of classes:", len(np.unique(y_train_full)))
print("Vocabulary size (num_words):", VOCAB_SIZE)

# Verify that labels are strictly binary (0 = negative, 1 = positive)
unique_labels = np.unique(np.concatenate([y_train_full, y_test]))
print("Unique labels found:", unique_labels)
assert set(unique_labels.tolist()) == {0, 1}, "Labels are not binary as expected!"

Downloading data from https://storage.googleapis.com/tensorflow/tf-
keras-datasets/imdb.npz
17464789/17464789 _____ 0s 0us/step
Training samples: 25000
Testing samples: 25000
Number of classes: 2
Vocabulary size (num_words): 10000
Unique labels found: [0 1]

```

4. Explore the Dataset

```

# Compute basic class-balance statistics
num_train = len(x_train_full)
num_test = len(x_test)
num_positive_train = int(np.sum(y_train_full == 1))
num_negative_train = int(np.sum(y_train_full == 0))

print(f"Number of training samples: {num_train}")
print(f"Number of test samples: {num_test}")
print(f"Number of positive reviews (train): {num_positive_train}")
print(f"Number of negative reviews (train): {num_negative_train}")

Number of training samples: 25000
Number of test samples: 25000
Number of positive reviews (train): 12500
Number of negative reviews (train): 12500

```

5. Analyze Sequence Lengths

```

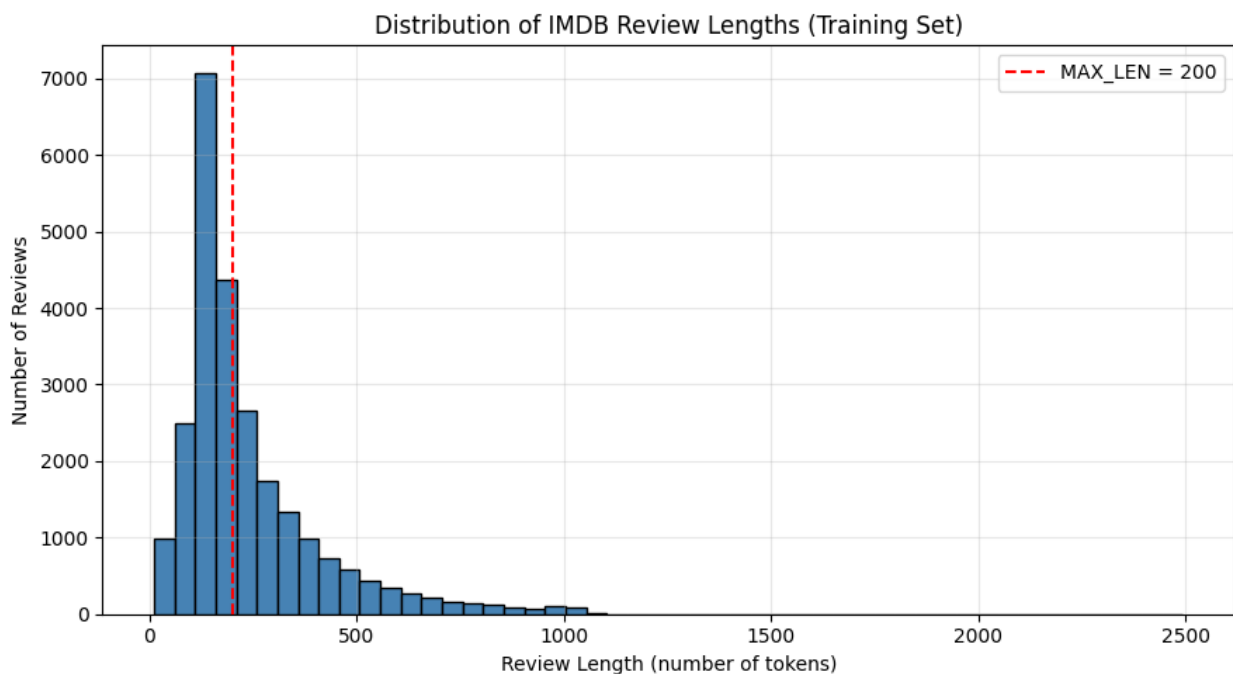
# Compute the length (number of tokens) of every training review
review_lengths = np.array([len(review) for review in x_train_full])

# Print summary statistics of review lengths
print("Minimum review length:", review_lengths.min())
print("Maximum review length:", review_lengths.max())
print("Average review length: {:.2f}".format(review_lengths.mean()))
print("Median review length:", int(np.median(review_lengths)))

```

```
# Plot the distribution of review lengths as a histogram
plt.figure(figsize=(9, 5))
plt.hist(review_lengths, bins=50, color="steelblue",
edgecolor="black")
plt.axvline(MAX_LEN, color="red", linestyle="--", label=f"MAX_LEN =
{MAX_LEN}")
plt.title("Distribution of IMDB Review Lengths (Training Set)")
plt.xlabel("Review Length (number of tokens)")
plt.ylabel("Number of Reviews")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

Minimum review length: 11
Maximum review length: 2494
Average review length: 238.71
Median review length: 178



6. Create Training and Validation Sets

```
# Split the 25,000 training samples into training and validation
subsets
# Stratify on labels to preserve class balance; the 25,000 test
samples are left untouched
x_train, x_val, y_train, y_val = train_test_split(
    x_train_full,
```

```

    y_train_full,
    test_size=5000,
    random_state=SEED,
    stratify=y_train_full
)

print("Training samples:", len(x_train))
print("Validation samples:", len(x_val))
print("Testing samples:", len(x_test))

```

```

Training samples: 20000
Validation samples: 5000
Testing samples: 25000

```

7. Pad/Truncate Sequences

```

# RNN/LSTM/GRU layers require fixed-length sequences for efficient batch processing
# Pad or truncate every review to exactly MAX_LEN tokens, applying padding/truncation at the end ("post")
x_train_padded = keras.utils.pad_sequences(x_train, maxlen=MAX_LEN, padding="post", truncating="post")
x_val_padded = keras.utils.pad_sequences(x_val, maxlen=MAX_LEN, padding="post", truncating="post")
x_test_padded = keras.utils.pad_sequences(x_test, maxlen=MAX_LEN, padding="post", truncating="post")

# Verify the resulting shapes before proceeding to training
print("x_train_padded shape:", x_train_padded.shape)
print("x_val_padded shape:", x_val_padded.shape)
print("x_test_padded shape:", x_test_padded.shape)

assert x_train_padded.shape == (20000, MAX_LEN)
assert x_val_padded.shape == (5000, MAX_LEN)
assert x_test_padded.shape == (25000, MAX_LEN)
print("All padded shapes match expected dimensions.")

x_train_padded shape: (20000, 200)
x_val_padded shape: (5000, 200)
x_test_padded shape: (25000, 200)
All padded shapes match expected dimensions.

```

8. Build Data Pipeline

```

# Build efficient tf.data pipelines for training, validation, and testing

# Training dataset: shuffled, batched, and prefetched for performance
train_dataset = (
    tf.data.Dataset.from_tensor_slices((x_train_padded, y_train))

```

```

        .shuffle(buffer_size=len(x_train_padded), seed=SEED)
        .batch(BATCH_SIZE)
        .prefetch(tf.data.AUTOTUNE)
    )

# Validation dataset: no shuffling needed, just batched and prefetched
    val_dataset = (
        tf.data.Dataset.from_tensor_slices((x_val_padded, y_val))
        .batch(BATCH_SIZE)
        .prefetch(tf.data.AUTOTUNE)
    )

# Test dataset: no shuffling, batched and prefetched
    test_dataset = (
        tf.data.Dataset.from_tensor_slices((x_test_padded, y_test))
        .batch(BATCH_SIZE)
        .prefetch(tf.data.AUTOTUNE)
    )

    print("Data pipelines created successfully.")

```

Data pipelines created successfully.

9. Model Architecture

All three models (RNN, LSTM, GRU) share the same overall architecture so that the comparison is fair:

```

Input -> Embedding -> Recurrent Layer (64 units) -> Dense(32, relu) ->
Dense(1, sigmoid)

```

Only the recurrent layer type differs between the three models. All models are compiled with the same optimizer, learning rate, loss function, and metrics, and trained for the same number of epochs with the same early-stopping configuration.

```

def build_model(recurrent_layer, name):
    """
        Build a sequence classification model using the given recurrent
        layer.

        Parameters
        -----
        recurrent_layer : a Keras recurrent layer instance (SimpleRNN,
        LSTM, or GRU)
        name : str, name to assign to the model

        Returns
        -----
        A compiled tf.keras.Sequential model.
    """

```

```

    """
    # Assemble the model: Embedding -> recurrent layer -> Dense ->
    Dense output
    model = Sequential([
        Embedding(input_dim=VOCAB_SIZE, output_dim=EMBEDDING_DIM), #
        recurrent_layer, #
        Dense(DENSE_UNITS, activation="relu"), #
        Dense(1, activation="sigmoid")
    ], name=name)

    # Compile with identical settings across all models for a fair
    comparison
    model.compile(
        optimizer=Adam(learning_rate=LEARNING_RATE),
        loss="binary_crossentropy",
        metrics=["accuracy"]
    )
    return model

def get_early_stopping():
    """Return a fresh EarlyStopping callback with the same
    configuration for every model."""
    return EarlyStopping(
        monitor="val_loss",
        patience=2,
        restore_best_weights=True
    )

def train_model(model, train_data, val_data, epochs=EPOCHS):
    """
    Train a model and record timing and history information.

    Returns
    -----
    history : the Keras History object
    training_time : float, total training time in seconds
    epochs_completed : int, actual number of epochs the model trained
    for
    """
    start_time = time.time()
    history = model.fit(
        train_data,
        validation_data=val_data,
        epochs=epochs,

```



```

        callbacks=[get_early_stopping()],
        verbose=1
    )
    training_time = time.time() - start_time
    epochs_completed = len(history.history["loss"])

    print(f"\n{model.name} training complete.")
    print(f"Epochs completed: {epochs_completed} (max allowed:
{epochs})")
    print(f"Final training accuracy:    {history.history['accuracy'][-
1]:.4f}")
    print(f"Final validation accuracy:
{history.history['val_accuracy'][-1]:.4f}")
    print(f"Final training loss:        {history.history['loss'][-
1]:.4f}")
    print(f"Final validation loss:      {history.history['val_loss'][-
1]:.4f}")
    print(f"Total training time: {training_time:.2f} seconds")

    return history, training_time, epochs_completed

def evaluate_model(model, x_test_data, y_test_data):
    """
    Evaluate a trained model on the test set.

    Returns a dictionary of test loss, accuracy, precision, recall,
    and F1-score.
    """
    # Get raw loss/accuracy from Keras' built-in evaluate
    test_loss, test_accuracy = model.evaluate(x_test_data,
y_test_data, verbose=0)

    # Get predicted probabilities and convert to binary predictions
    using a 0.5 threshold
    probabilities = model.predict(x_test_data, verbose=0).ravel()
    predictions = (probabilities >= 0.5).astype(int)

    precision = precision_score(y_test_data, predictions)
    recall = recall_score(y_test_data, predictions)
    f1 = f1_score(y_test_data, predictions)

    return {
        "test_loss": test_loss,
        "test_accuracy": test_accuracy,
        "precision": precision,
        "recall": recall,
        "f1_score": f1,
        "predictions": predictions,
        "probabilities": probabilities
    }

```

```

    }

    print("Helper functions defined: build_model, get_early_stopping,
          train_model, evaluate_model")

```

Helper functions defined: build_model, get_early_stopping, train_model, evaluate_model

10. Build, Train, and Evaluate the RNN Model

```

# Build the SimpleRNN model using the shared architecture
rnn_model = build_model(SimpleRNN(RNN_UNITS), name="SimpleRNN")
rnn_model.summary()

```

Model: "SimpleRNN"

Layer (type) Param #	Output Shape	
embedding (Embedding) (unbuilt)	?	0
simple_rnn (SimpleRNN) (unbuilt)	?	0
dense (Dense) (unbuilt)	?	0
dense_1 (Dense) (unbuilt)	?	0

Total params: 0 (0.00 B)

Trainable params: 0 (0.00 B)

Non-trainable params: 0 (0.00 B)

```

# Train the RNN model

```

```

rnn_history, rnn_training_time, rnn_epochs_completed =
train_model(rnn_model, train_dataset, val_dataset)

```

Epoch 1/5

313/313 ————— 24s 69ms/step - accuracy: 0.5013 - loss:

```

0.6972 - val_accuracy: 0.5044 - val_loss: 0.6958
Epoch 2/5
313/313 _____ 22s 69ms/step - accuracy: 0.5020 - loss:
0.6974 - val_accuracy: 0.5002 - val_loss: 0.6968
Epoch 3/5
313/313 _____ 41s 69ms/step - accuracy: 0.5153 - loss:
0.6930 - val_accuracy: 0.5054 - val_loss: 0.6927
Epoch 4/5
313/313 _____ 22s 71ms/step - accuracy: 0.5642 - loss:
0.6742 - val_accuracy: 0.5094 - val_loss: 0.7042
Epoch 5/5
313/313 _____ 40s 69ms/step - accuracy: 0.5976 - loss:
0.6287 - val_accuracy: 0.5092 - val_loss: 0.7286

```

```

SimpleRNN training complete.
Epochs completed: 5 (max allowed: 5)
Final training accuracy: 0.5976
Final validation accuracy: 0.5092
Final training loss: 0.6287
Final validation loss: 0.7286
Total training time: 148.82 seconds

```

```

# Evaluate the RNN model on the untouched test set
rnn_results = evaluate_model(rnn_model, x_test_padded, y_test)

```

```

print("RNN Test Results:")
print(f" Test Loss: {rnn_results['test_loss']:.4f}")
print(f" Test Accuracy: {rnn_results['test_accuracy']:.4f}")
print(f" Precision: {rnn_results['precision']:.4f}")
print(f" Recall: {rnn_results['recall']:.4f}")
print(f" F1-score: {rnn_results['f1_score']:.4f}")

```

```

RNN Test Results:
Test Loss: 0.6937
Test Accuracy: 0.5051
Precision: 0.5028
Recall: 0.9334
F1-score: 0.6535

```

11. Build, Train, and Evaluate the LSTM Model

```

# Build the LSTM model using the same embedding and dense structure as
the RNN

```

```

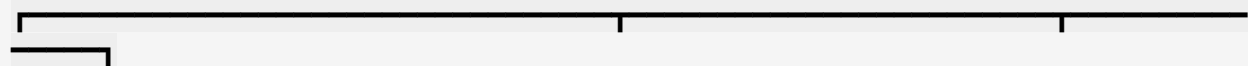
lstm_model = build_model(LSTM(RNN_UNITS), name="LSTM")
lstm_model.summary()

```

```

Model: "LSTM"

```



Layer (type) Param #	Output Shape	
embedding_1 (Embedding) (unbuilt)	?	0
lstm (LSTM) (unbuilt)	?	0
dense_2 (Dense) (unbuilt)	?	0
dense_3 (Dense) (unbuilt)	?	0

Total params: 0 (0.00 B)

Trainable params: 0 (0.00 B)

Non-trainable params: 0 (0.00 B)

Train the LSTM model with identical settings to the RNN

lstm_history, lstm_training_time, lstm_epochs_completed =
train_model(lstm_model, train_dataset, val_dataset)

Epoch 1/5

313/313 ————— 48s 146ms/step - accuracy: 0.5310 - loss:
0.6862 - val_accuracy: 0.5206 - val_loss: 0.6842

Epoch 2/5

313/313 ————— 43s 139ms/step - accuracy: 0.5741 - loss:
0.6605 - val_accuracy: 0.6070 - val_loss: 0.6342

Epoch 3/5

313/313 ————— 46s 146ms/step - accuracy: 0.6847 - loss:
0.5761 - val_accuracy: 0.5054 - val_loss: 0.8212

Epoch 4/5

313/313 ————— 44s 140ms/step - accuracy: 0.6181 - loss:
0.6405 - val_accuracy: 0.7474 - val_loss: 0.5778

Epoch 5/5

313/313 ————— 44s 140ms/step - accuracy: 0.7238 - loss:
0.5359 - val_accuracy: 0.6888 - val_loss: 0.6264

LSTM training complete.

Epochs completed: 5 (max allowed: 5)

Final training accuracy: 0.7238

```

Final validation accuracy: 0.6888
Final training loss:      0.5359
Final validation loss:    0.6264
Total training time: 224.67 seconds

# Evaluate the LSTM model on the untouched test set
lstm_results = evaluate_model(lstm_model, x_test_padded, y_test)

print("LSTM Test Results:")
print(f"  Test Loss:      {lstm_results['test_loss']:.4f}")
print(f"  Test Accuracy:   {lstm_results['test_accuracy']:.4f}")
print(f"  Precision:       {lstm_results['precision']:.4f}")
print(f"  Recall:          {lstm_results['recall']:.4f}")
print(f"  F1-score:        {lstm_results['f1_score']:.4f}")

LSTM Test Results:
  Test Loss:      0.5885
  Test Accuracy:  0.7348
  Precision:      0.7393
  Recall:         0.7254
  F1-score:       0.7323

```

12. Build, Train, and Evaluate the GRU Model

```

# Build the GRU model using the same embedding and dense structure as
the RNN and LSTM
gru_model = build_model(GRU(RNN_UNITS), name="GRU")
gru_model.summary()

```

Model: "GRU"

Layer (type) Param #	Output Shape	
embedding_2 (Embedding) (unbuilt)	?	0
gru (GRU) (unbuilt)	?	0
dense_4 (Dense) (unbuilt)	?	0
dense_5 (Dense)	?	0

```
(unbuilt) |
```

```
Total params: 0 (0.00 B)
```

```
Trainable params: 0 (0.00 B)
```

```
Non-trainable params: 0 (0.00 B)
```

```
# Train the GRU model with identical settings to the RNN and LSTM
```

```
gru_history, gru_training_time, gru_epochs_completed =  
train_model(gru_model, train_dataset, val_dataset)
```

```
Epoch 1/5
```

```
313/313 _____ 56s 170ms/step - accuracy: 0.5105 - loss:  
0.6932 - val_accuracy: 0.5190 - val_loss: 0.6908
```

```
Epoch 2/5
```

```
313/313 _____ 53s 169ms/step - accuracy: 0.6661 - loss:  
0.5754 - val_accuracy: 0.8270 - val_loss: 0.4205
```

```
Epoch 3/5
```

```
313/313 _____ 53s 169ms/step - accuracy: 0.8882 - loss:  
0.2785 - val_accuracy: 0.8592 - val_loss: 0.3475
```

```
Epoch 4/5
```

```
313/313 _____ 53s 169ms/step - accuracy: 0.9386 - loss:  
0.1722 - val_accuracy: 0.8648 - val_loss: 0.3795
```

```
Epoch 5/5
```

```
313/313 _____ 92s 200ms/step - accuracy: 0.9654 - loss:  
0.1005 - val_accuracy: 0.8552 - val_loss: 0.4706
```

```
GRU training complete.
```

```
Epochs completed: 5 (max allowed: 5)
```

```
Final training accuracy: 0.9654
```

```
Final validation accuracy: 0.8552
```

```
Final training loss: 0.1005
```

```
Final validation loss: 0.4706
```

```
Total training time: 306.23 seconds
```

```
# Evaluate the GRU model on the untouched test set
```

```
gru_results = evaluate_model(gru_model, x_test_padded, y_test)
```

```
print("GRU Test Results:")
```

```
print(f" Test Loss: {gru_results['test_loss']:.4f}")
```

```
print(f" Test Accuracy: {gru_results['test_accuracy']:.4f}")
```

```
print(f" Precision: {gru_results['precision']:.4f}")
```

```
print(f" Recall: {gru_results['recall']:.4f}")
```

```
print(f" F1-score: {gru_results['f1_score']:.4f}")
```

```
GRU Test Results:
```

```
Test Loss: 0.3411
```

```
Test Accuracy: 0.8558
```

Precision:	0.8392
Recall:	0.8802
F1-score:	0.8592

13. Compare All Models

```
# Collect model names, histories, and training times for easy
comparison
model_names = ["RNN", "LSTM", "GRU"]
histories = {"RNN": rnn_history, "LSTM": lstm_history, "GRU":
gru_history}
training_times = {"RNN": rnn_training_time, "LSTM":
lstm_training_time, "GRU": gru_training_time}
epochs_completed_dict = {"RNN": rnn_epochs_completed, "LSTM":
lstm_epochs_completed, "GRU": gru_epochs_completed}
results_dict = {"RNN": rnn_results, "LSTM": lstm_results, "GRU":
gru_results}
models_dict = {"RNN": rnn_model, "LSTM": lstm_model, "GRU": gru_model}

print("Epochs completed per model (early stopping may reduce this
below EPOCHS):")
for name in model_names:
    print(f" {name}: {epochs_completed_dict[name]} / {EPOCHS}")

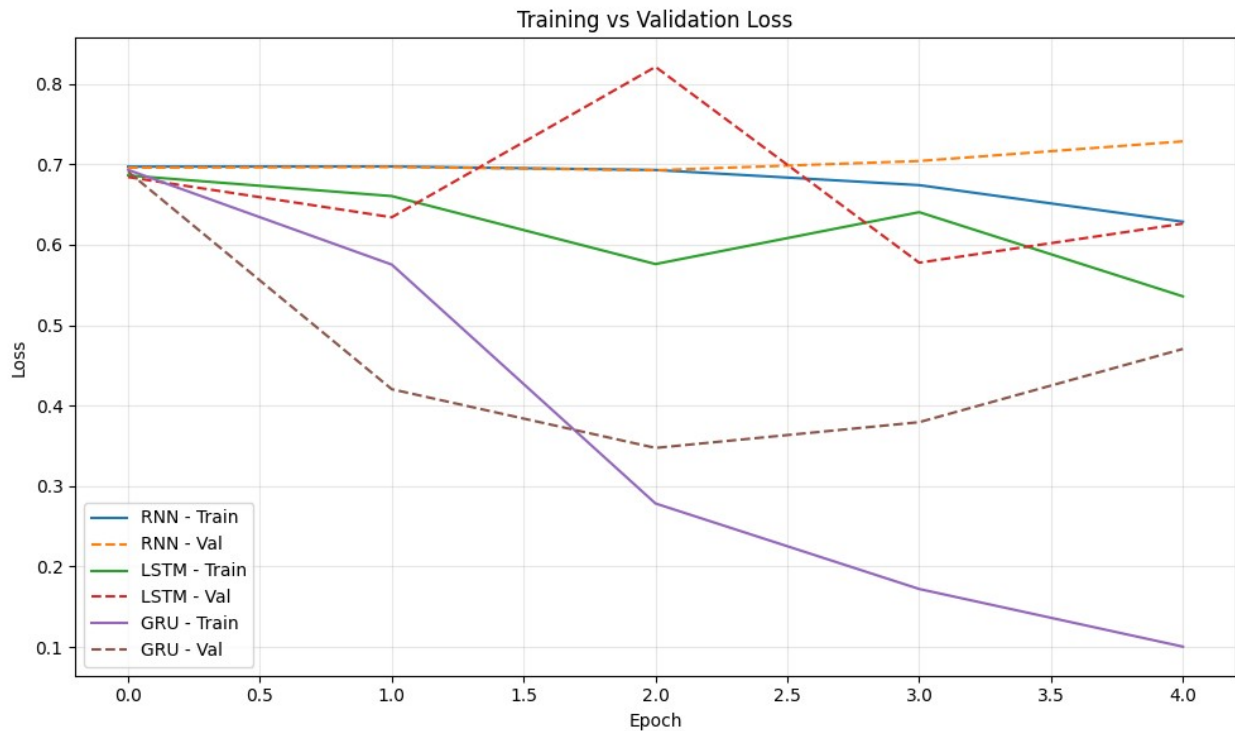
Epochs completed per model (early stopping may reduce this below
EPOCHS):
RNN: 5 / 5
LSTM: 5 / 5
GRU: 5 / 5
```

14. Plot Training and Validation Curves

```
# Compare training and validation accuracy across all three models
plt.figure(figsize=(10, 6))
for name in model_names:
    plt.plot(histories[name].history["accuracy"], label=f"{name} -
Train")
    plt.plot(histories[name].history["val_accuracy"], linestyle="--",
label=f"{name} - Val")
plt.title("Training vs Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
# Compare training and validation loss across all three models
plt.figure(figsize=(10, 6))
for name in model_names:
    plt.plot(histories[name].history["loss"], label=f"{name} - Train")
    plt.plot(histories[name].history["val_loss"], linestyle="--",
label=f"{name} - Val")
plt.title("Training vs Validation Loss")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.legend()
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

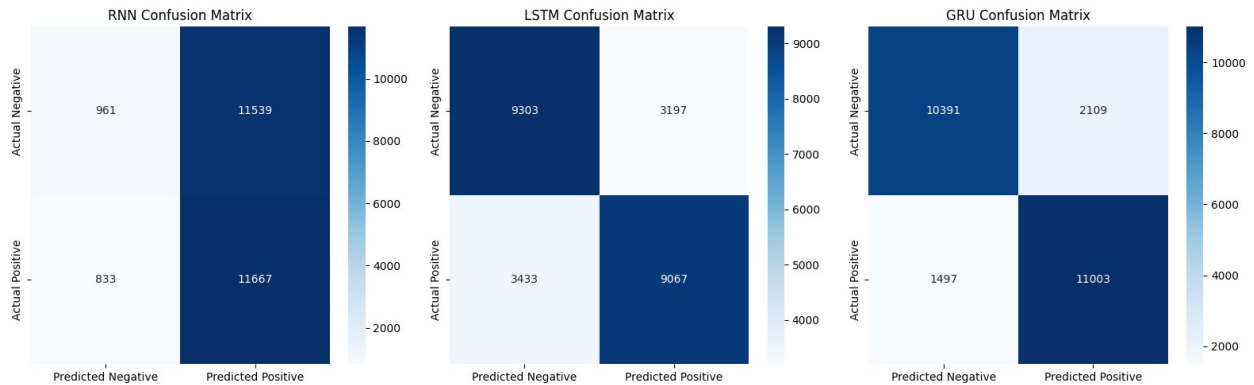



15. Confusion Matrices

```
# Plot a confusion matrix heatmap for each model
fig, axes = plt.subplots(1, 3, figsize=(16, 5))

for ax, name in zip(axes, model_names):
    cm = confusion_matrix(y_test, results_dict[name]["predictions"])
    sns.heatmap(
        cm, annot=True, fmt="d", cmap="Blues", ax=ax,
        xticklabels=["Predicted Negative", "Predicted Positive"],
        yticklabels=["Actual Negative", "Actual Positive"]
    )
    ax.set_title(f"{name} Confusion Matrix")

plt.tight_layout()
plt.show()
```



16. Classification Reports

Print a classification report for each model

```
for name in model_names:
    print(f"\n==== {name} Classification Report =====")
    print(classification_report(
        y_test,
        results_dict[name]["predictions"],
        target_names=["Negative", "Positive"]
    ))
```

==== RNN Classification Report =====

	precision	recall	f1-score	support
Negative	0.54	0.08	0.13	12500
Positive	0.50	0.93	0.65	12500
accuracy			0.51	25000
macro avg	0.52	0.51	0.39	25000
weighted avg	0.52	0.51	0.39	25000

==== LSTM Classification Report =====

	precision	recall	f1-score	support
Negative	0.73	0.74	0.74	12500
Positive	0.74	0.73	0.73	12500
accuracy			0.73	25000
macro avg	0.73	0.73	0.73	25000
weighted avg	0.73	0.73	0.73	25000

==== GRU Classification Report =====

	precision	recall	f1-score	support
Negative	0.87	0.83	0.85	12500
Positive	0.84	0.88	0.86	12500

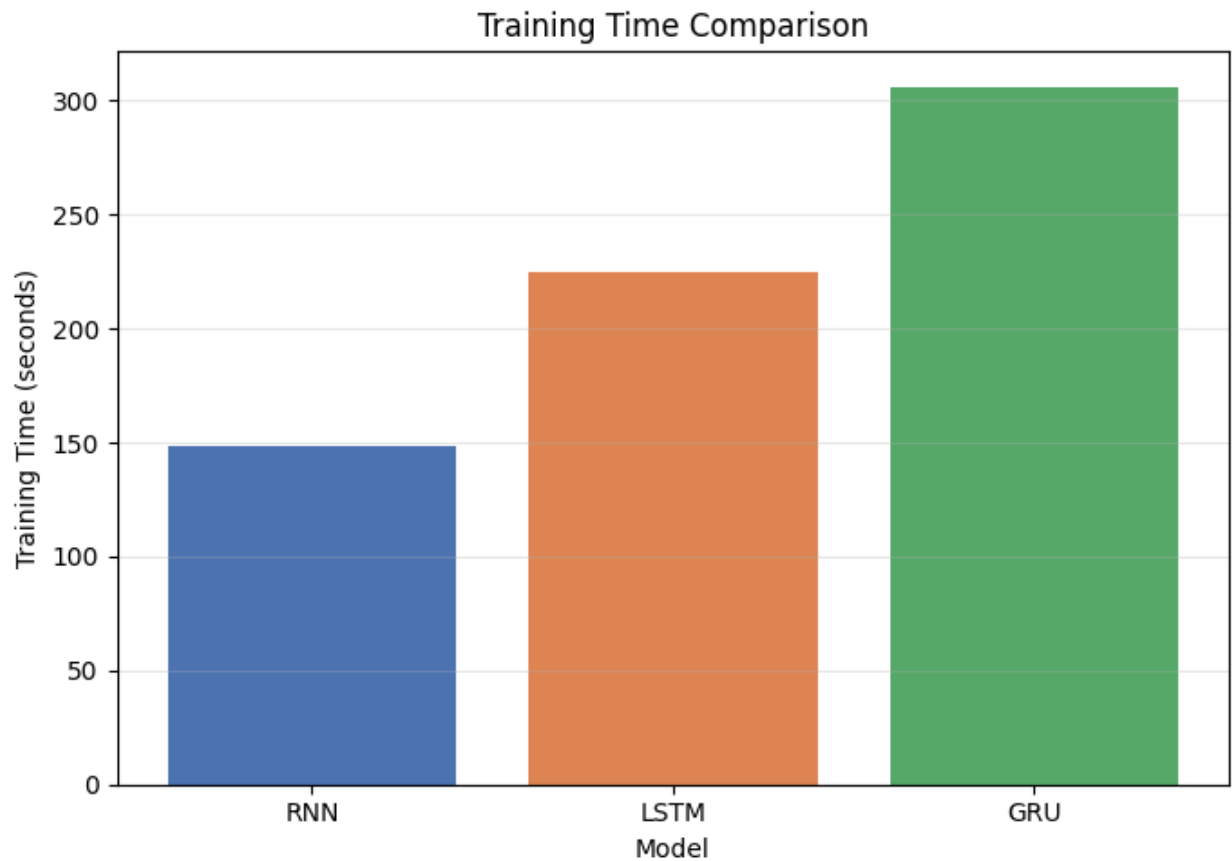
accuracy			0.86	25000
macro avg	0.86	0.86	0.86	25000
weighted avg	0.86	0.86	0.86	25000

17. Training Time Comparison

```
# Print exact training times
for name in model_names:
    print(f"{name} training time: {training_times[name]:.2f} seconds
({epochs_completed_dict[name]} epochs)")
```

```
# Bar chart comparing training time across models
plt.figure(figsize=(7, 5))
plt.bar(model_names, [training_times[n] for n in model_names],
        color=["#4C72B0", "#DD8452", "#55A868"])
plt.title("Training Time Comparison")
plt.xlabel("Model")
plt.ylabel("Training Time (seconds)")
plt.grid(alpha=0.3, axis="y")
plt.tight_layout()
plt.show()
```

```
RNN training time: 148.82 seconds (5 epochs)
LSTM training time: 224.67 seconds (5 epochs)
GRU training time: 306.23 seconds (5 epochs)
```



18. Parameter Comparison

```
# Compute total and trainable parameter counts for each model
param_rows = []
for name in model_names:
    model = models_dict[name]
    total_params = model.count_params()
    trainable_params = int(np.sum([np.prod(w.shape) for w in
model.trainable_weights]))
    param_rows.append({
        "Model": name,
        "Total Parameters": total_params,
        "Trainable Parameters": trainable_params
    })

param_df = pd.DataFrame(param_rows)
print(param_df)
```

	Model	Total Parameters	Trainable Parameters
0	RNN	650369	650369
1	LSTM	675137	675137
2	GRU	667073	667073

19. Final Comparison Table

```
# Build the final comparison DataFrame, one row per model
comparison_rows = []
for name in model_names:
    r = results_dict[name]
    comparison_rows.append({
        "Model": name,
        "Test Loss": r["test_loss"],
        "Test Accuracy": r["test_accuracy"],
        "Precision": r["precision"],
        "Recall": r["recall"],
        "F1 Score": r["f1_score"],
        "Training Time (seconds)": training_times[name],
        "Total Parameters": param_df.loc[param_df["Model"] == name,
        "Total Parameters"].values[0],
        "Trainable Parameters": param_df.loc[param_df["Model"] ==
name, "Trainable Parameters"].values[0]
    })

comparison_df = pd.DataFrame(comparison_rows)

# Sort by Test Accuracy in descending order
comparison_df = comparison_df.sort_values(by="Test Accuracy",
ascending=False).reset_index(drop=True)

print(comparison_df)

# Save the comparison table to a CSV file
comparison_df.to_csv("rnn_lstm_gru_comparison.csv", index=False)
print("\nSaved comparison table to rnn_lstm_gru_comparison.csv")
```

	Model	Test Loss	Test Accuracy	Precision	Recall	F1 Score	\
0	GRU	0.341117	0.85576	0.839155	0.88024	0.859207	
1	LSTM	0.588484	0.73480	0.739318	0.72536	0.732273	
2	RNN	0.693707	0.50512	0.502758	0.93336	0.653504	

	Training Time (seconds)	Total Parameters	Trainable Parameters
0	306.226166	667073	667073
1	224.672240	675137	675137
2	148.815237	650369	650369

Saved comparison table to rnn_lstm_gru_comparison.csv

20. Best Model

```
# Automatically identify the best-performing model on each metric (no
hard-coded values)
best_accuracy_model = comparison_df.loc[comparison_df["Test
Accuracy"].idxmax(), "Model"]
```

```

best_f1_model = comparison_df.loc[comparison_df["F1 Score"].idxmax(),
"Model"]
fastest_model = comparison_df.loc[comparison_df["Training Time
(seconds)"].idxmin(), "Model"]

print(f"Best Model by Accuracy: {best_accuracy_model}")
print(f"Best Model by F1 Score: {best_f1_model}")
print(f"Fastest Model: {fastest_model}")

Best Model by Accuracy: GRU
Best Model by F1 Score: GRU
Fastest Model: RNN

```

21. Final Analysis

```

# Dynamically generate a final analysis based on the actual results in
comparison_df
best_row = comparison_df.iloc[0]
fastest_row = comparison_df.loc[comparison_df["Training Time
(seconds)"].idxmin()]

analysis = f"""
FINAL ANALYSIS
=====

1. Best performing model (by test accuracy): {best_accuracy_model}
   - Test Accuracy: {comparison_df.loc[comparison_df['Model'] ==
best_accuracy_model, 'Test Accuracy'].values[0]:.4f}

2. Fastest model to train: {fastest_model}
   - Training Time: {fastest_row['Training Time (seconds)']:.2f}
seconds

3. Best F1-score: {best_f1_model}
   - F1 Score: {comparison_df.loc[comparison_df['Model'] ==
best_f1_model, 'F1 Score'].values[0]:.4f}

print(analysis)

FINAL ANALYSIS
=====

1. Best performing model (by test accuracy): GRU
   - Test Accuracy: 0.8558

2. Fastest model to train: RNN
   - Training Time: 148.82 seconds

```

3. Best F1-score: GRU

- F1 Score: 0.8592

4. How the architectures differ:

- SimpleRNN passes a hidden state forward through a single tanh transformation at every timestep, with no dedicated mechanism to control what information is kept or discarded.
- LSTM introduces input, forget, and output gates plus a separate cell state, giving it explicit control over what information is remembered or forgotten over long sequences.
- GRU simplifies the LSTM by merging the cell and hidden state and using only two gates (update and reset), reducing the number of parameters while still controlling information flow across timesteps.

5. Why LSTM and GRU handle long-term dependencies better than a basic RNN:

Their gating mechanisms allow gradients to flow through many timesteps largely unchanged when the gates are open, which directly mitigates the vanishing gradient problem that plain RNNs suffer from over long sequences.

6. Why GRU may train faster than LSTM:

GRU has fewer gates and no separate cell state, meaning fewer parameters and fewer matrix operations per timestep than LSTM, which typically translates to faster training and inference for a similar hidden size.

7. Why basic RNNs may suffer from vanishing gradients:

During backpropagation through time, repeated multiplication by the same weight matrix and the derivative of the tanh activation causes gradients to shrink exponentially over long sequences, making it hard for a plain RNN to learn dependencies that span many timesteps.

8. Accuracy vs computational cost trade-off:

Based on the measured results above, the model with the highest test accuracy (GRU) required 306.23 seconds to train, while the fastest model (RNN) trained in 148.82 seconds. Whether the extra accuracy is worth

the added training cost depends on the deployment constraints - for latency-sensitive or resource-constrained settings, the faster model may be preferable even at a small accuracy cost; for offline or accuracy-critical applications, the most accurate model is usually the better choice.